PROJECT GUIDE · 2026-11-02

ecg-trust

A 12-lead ECG classifier that ships with evidence for *when it can be trusted*: patient-level evaluation, calibration, calibrated abstention, subgroup and noise robustness, external validation, and a comparison with a commercial algorithm.

PTB-XL · 21,799 ECGs | External · 28,646 ECGs | PyTorch · CPU | Conformal prediction

0.925
macro-AUROC, internal test (95% CI 0.917–0.932)

0.949
macro-AUROC, external validation (95% CI 0.946–0.951)

0.033 → 0.030
macro ECE on test, before → after temperature scaling

13%
ECGs referred at α = 0.10 (internal test)

Author: Rittik Patra. Research software — **not a medical device** and not for diagnostic use. All numbers in this guide are read from the saved, version-controlled reports in `ecg-trust/reports/`; none are typed by hand.

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1 · Executive summary

Most ECG deep-learning portfolios stop at one number: the AUROC on a held-out split. A clinical team deciding whether to deploy a model needs more: are the probabilities honest, can the model say "I don't know", does it hold up on noisy recordings, on other hospitals and on patient subgroups, and is it better than the interpretation software already built into the ECG cart? This project answers those questions for a five-class 12-lead ECG classifier trained on PTB-XL.

> **Headline.** On the locked PTB-XL test fold the 3-seed ResNet ensemble reaches macro-AUROC 0.925 (95% CI 0.917–0.932); temperature scaling moves macro ECE 0.033 → 0.030; at α = 0.10 about 13% of label decisions are referred. Fusing the ensemble with GE 12SL measurements gives 0.939 (95% CI 0.933–0.945). On the external database macro-AUROC is 0.949 (95% CI 0.946–0.951).

Question	How it is answered	Result
Does it rank cases well?	Patient-bootstrap AUROC, internal test	0.925
Are probabilities honest?	ECE before → after temperature scaling	0.033 → 0.030
Can it say 'I don't know'?	Label-conditional conformal refer rate, α = 0.10	13%
Does adding 12SL measurements help?	ResNet + 12SL logit fusion, internal test	0.939
Does it transfer?	External database, strict label map	0.949
Is it robust to noise?	Synthetic noise at exact SNR; refer rate vs SNR	see §6.4
Better than the cart software?	Sensitivity at GE 12SL's specificity	see §6.7

2 · Objective and motivation

Objective. Build, evaluate and serve a classifier for the five PTB-XL diagnostic superclasses — normal ECG (NORM), myocardial infarction (MI), ST/T change (STTC), conduction disturbance (CD) and hypertrophy (HYP) — from 10-second, 12-lead ECGs, and produce evidence for when its output can be trusted.

Why trust is the product. A diagnostic-support model is only useful if a clinician can act on its output. Three properties matter as much as ranking performance. *Calibration*: a predicted 70 % should be right about 70 % of the time, otherwise thresholds and risk communication break. *Abstention*: on ambiguous or poor-quality recordings the model should refer to a cardiologist instead of guessing. *Transportability*: performance on a different population, device and labelling process is usually lower; the size of the drop must be measured, not assumed.

Scope. Multi-label classification at 100 Hz. The 500 Hz recordings, the 71 individual statements, new architectures and clinical thresholds are out of scope. Compute is a 4-core CPU container, so models are deliberately compact.

What a reader should take from this project

Data rigour
Checksum-verified downloads, audited labels, patient-level splits with an automated leakage test.

Honest evaluation
One-shot, locked test set; patient-bootstrap CIs; an external dataset from another country.

Trust layer
Temperature scaling and per-class conformal "present / absent / refer" decisions.

Engineering
Tested package, CLI, CI, FastAPI service, Streamlit demo, Docker, model card.

3 · Plan and decision log

The project followed a written design specification and a 24-task, test-driven implementation plan with four milestones. Each milestone ended at a review gate with the author.

Milestone	Deliverable	Gate
M1 · Data	Downloader, labels, splits, signal caches, external loader, feature baseline, data audit	Audit and label-map review
M2 · Models	1-D ResNet, 3-seed ensemble, calibration, conformal, robustness, 12SL comparator — validation data only	Freeze: models, thresholds and maps hashed
M3 · Final evaluation	Single logged run on the internal test fold and on the external set	Results review
M4 · Product	API, demo, Docker, CI, model card, README, this guide	Portfolio-ready

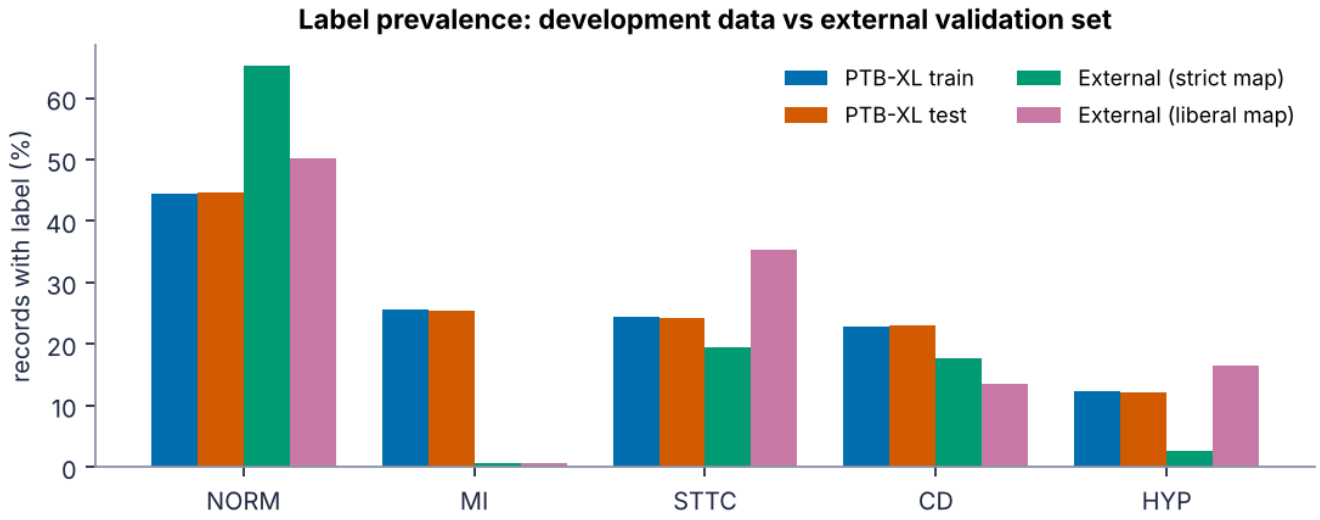
Decisions taken during the project

Decision	Why
PhysioNet AWS open-data mirror, every file SHA-256-verified, resumable downloads	physionet.org was not reachable; the proxy cut long transfers
Patient-level split of fold 9 into model-selection and calibration halves	conformal guarantees need calibration data unused for training or tuning
External label map completed to all 94 SNOMED codes; strict PTB-XL-faithful primary map, liberal map as sensitivity analysis	the dataset's condition list missed 43 codes, incl. complete RBBB and LBBB; PTB-XL treats isolated ST/T findings as form statements
NORM proxy = sinus-family rhythms only; unlabelled external records excluded	mirrors PTB-XL, where sinus bradycardia can be a normal ECG
1-D ResNet width 64 (1.49 M parameters), 3 seeds	author's choice after a CPU timing benchmark
Synthetic, SNR-exact noise instead of NSTDB records	NSTDB was not on the reachable mirror
GE 12SL map built from statement names cross-checked with PTB-XL+ SNOMED concepts; ambiguous statements unmapped	keeps the commercial comparison conservative
Patience raised to 30 epochs (no early stop); best-validation epoch kept	with a one-cycle schedule, patience 5 stopped at the learning-rate peak
Late logit fusion of the ResNet ensemble with the 12SL baseline (w = 0.4 on val_select)	at the freeze gate the ensemble (0.930) trailed the 12SL features (0.939) on validation
Freeze manifest + single-use test log per model	prevents tuning on the test sets

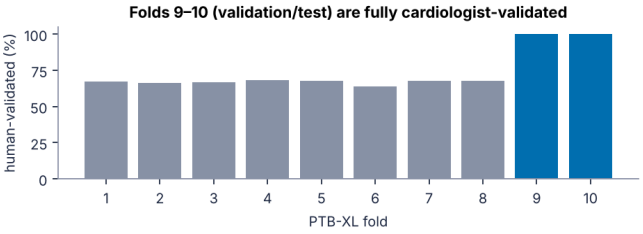
4 · Data

Dataset	Role	Size
PTB-XL 1.0.3 (Wagner et al., 2020)	Training, validation, internal test	21,799 ECGs, 18,869 patients, 10 s, 12 leads
PTB-XL+ 1.0.1 (Strodthoff et al., 2023)	Feature baseline; GE 12SL machine statements	ECG measurements from GE 12SL, Glasgow Uni-G and ECGDeli
Large-scale 12-lead arrhythmia database 1.0.0 (Zheng et al., 2022)	External validation	45,152 ECGs from Chapman University / Shaoxing / Ningbo hospitals; 28,646 evaluated

All files were downloaded from the PhysioNet open-data mirror and verified against the published SHA-256 checksums (134,363 files, 6.2 GB, zero failures). Labels follow the published benchmark convention: a superclass is positive if any diagnostic SCP statement maps to it. 411 PTB-XL records have no diagnostic superclass and are excluded.



Label prevalence. The external set is dominated by sinus-rhythm-only ECGs (the NORM proxy) and has very few MI labels — a large, deliberate dataset shift.



Share of labels validated by a cardiologist per PTB-XL fold. Evaluation folds are cleaner than training folds.

Splits (patient-level)

train	17,084 records
val_select	1,072 records
val_calib	1,074 records
test	2,158 records

Folds 1–8 train; fold 9 split by patient into model selection and calibration halves; fold 10 is the locked test set. No patient appears in two splits (checked by a unit test and at run time).

Harmonising the external labels

The external database codes diagnoses in SNOMED CT; 43 of the 94 codes present in its records were missing from its own condition list and were identified from the PhysioNet/CinC Challenge 2021 mapping. The primary map is strict and PTB-XL-faithful: isolated ST/T-wave findings and voltage-only LVH criteria are *form* statements in PTB-XL and are therefore not mapped. A liberal map that includes them is reported as a sensitivity analysis. NORM is a proxy (sinus-family rhythm codes only), and records without any superclass are excluded, exactly as in PTB-XL.

5 · Methods

1 · Signals
100 Hz, 12 × 1000, mV; per-lead standardisation fitted on train only.

2 · Models
Gradient boosting on PTB-XL+ features; compact 1-D ResNet × 3 seeds.

3 · Calibration
Per-class temperature fitted on the calibration half of fold 9.

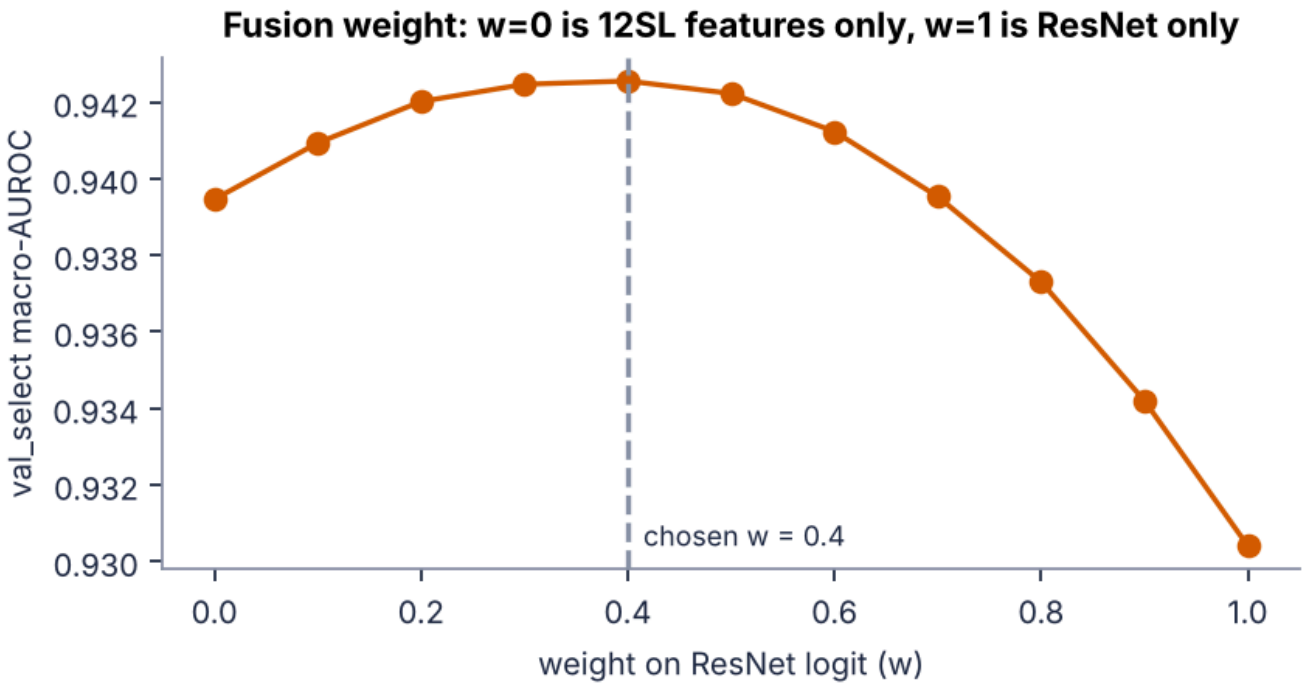
4 · Abstention
Per-class Mondrian conformal sets → present / absent / refer.

5 · Evaluation
Locked one-shot test and external runs, bootstrap CIs, subgroups, noise.

Models

Feature baseline. One histogram gradient-boosting classifier per class on 782 GE 12SL measurements (intervals, amplitudes, axes), with ECGDeli features as an open-source alternative. **Deep model.** A 1-D ResNet (width 64, stages [2, 2, 2, 2], kernel 7) trained with AdamW and a one-cycle learning rate for 30 epochs, keeping the epoch with the best validation macro-AUROC; three seeds are averaged as a deep ensemble (Lakshminarayanan et al., 2017).

Fusion. The ensemble's probability and the 12SL baseline's probability are combined in logit space, $z = w \cdot \mathrm{logit}(p_{\mathrm{ResNet}}) + (1 - w) \cdot \mathrm{logit}(p_{\mathrm{12SL}})$, with one weight w = 0.4 chosen on val_select from an 11-point grid. It is calibrated and given conformal thresholds on val_calib exactly like the ResNet. Because it needs GE 12SL measurements, it is evaluated on PTB-XL only.



Fusion weight chosen on validation only: w = 0 is the 12SL feature model, w = 1 the ResNet ensemble.

Calibration

For each class a single temperature T rescales every member's logit before averaging: $p = \mathrm{mean}_m\ \sigma(z_m / T)$, with T chosen to minimise the negative log-likelihood on the calibration split (Guo et al., 2017). Calibration is measured with the expected calibration error (15 equal-mass bins), Brier score and reliability diagrams.

Conformal abstention

For each class, non-conformity scores are 1 − p for positive and p for negative calibration ECGs, and each label value gets its own finite-sample quantile (label-conditional, "Mondrian" split conformal; Vovk et al., 2005; Angelopoulos & Bates, 2021):

$$\hat{q}_y = \text{the } \lceil (n_y + 1)(1 - \alpha) \rceil \text{-th smallest score among calibration ECGs with label } y$$

A new ECG's set contains "present" if $1 - p \le \hat{q}_1$ and "absent" if $p \le \hat{q}_0$. A single-element set is a decision; otherwise the ECG is referred. Under exchangeability the true label is covered with probability ≥ $1 - \alpha$ separately for positives and negatives.

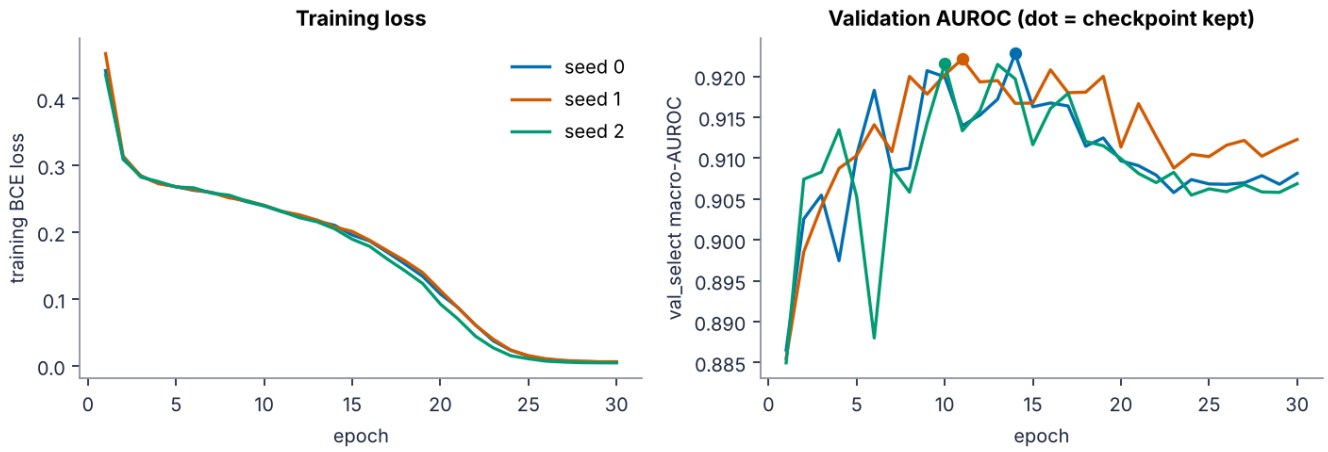
> **Caveat.** The guarantee is marginal and assumes new data look like the calibration data. The external set violates this on purpose; its coverage is reported to show by how much.

Robustness, subgroups and comparator

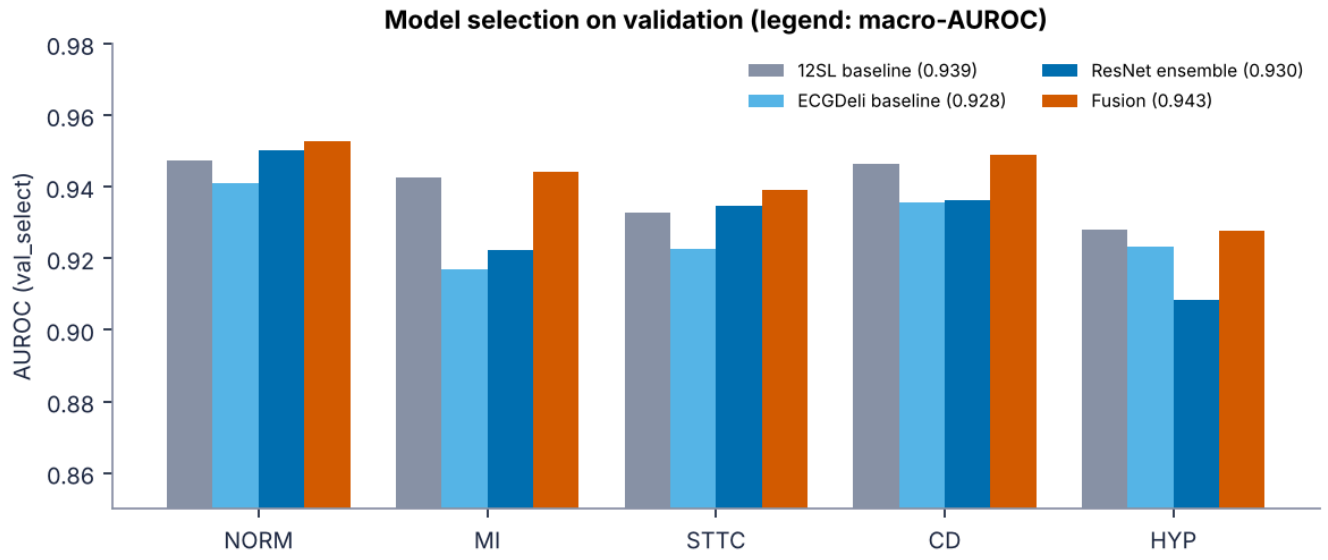
Synthetic noise mimicking the MIT-BIH Noise Stress Test types (baseline wander below 0.5 Hz, EMG-like 20–45 Hz, electrode-motion steps and bursts) is added at exact per-lead SNRs from 24 to −6 dB, plus lead dropout. Subgroups (sex, age band, device, PTB-XL noise flags, pacemaker) are descriptive, with patient-bootstrap CIs; groups with fewer than 30 positives or negatives are marked insufficient. The GE 12SL comparison uses 12SL's own operating point: the model's sensitivity is read at 12SL's specificity on the same ECGs, using a published statement-to-superclass map.

6 · Results

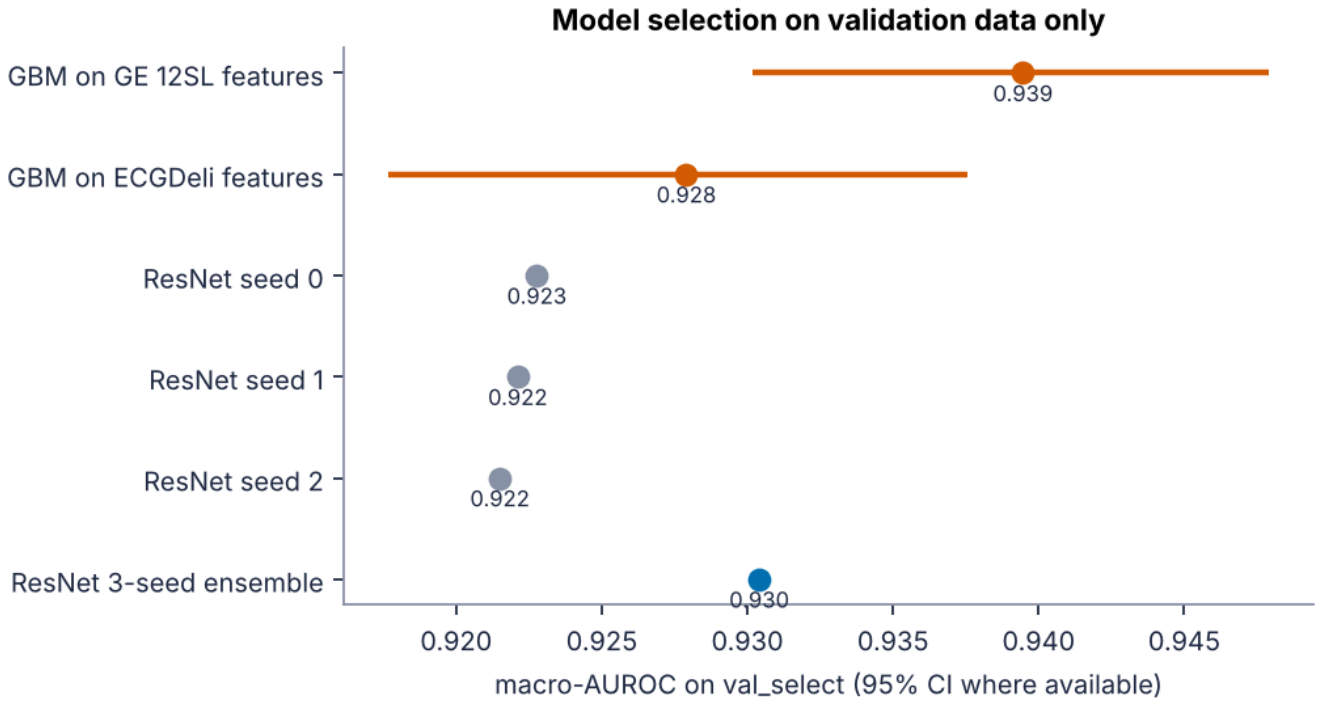
6.1 Training and model selection (validation data only)



Training loss and validation macro-AUROC per epoch for each seed; the dot marks the epoch kept. Validation AUROC peaks around epochs 10–15 while training loss keeps falling, so the best-validation checkpoint matters.



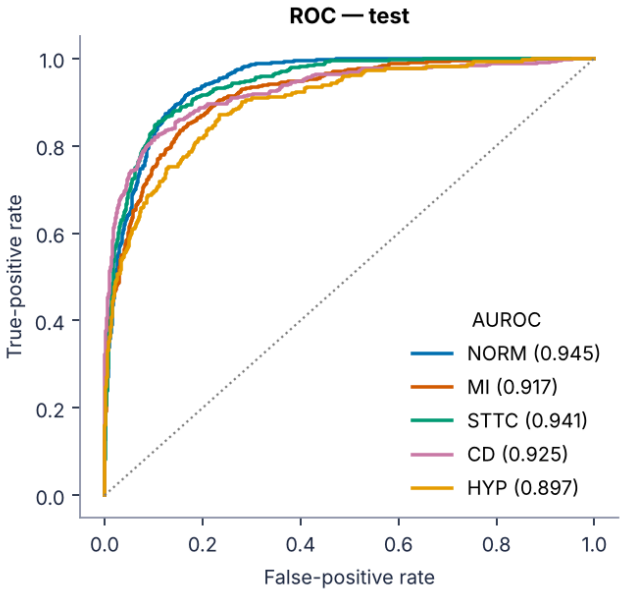
Per-class validation AUROC for every candidate model. The fused model is best overall; the ResNet alone trails the 12SL features slightly.



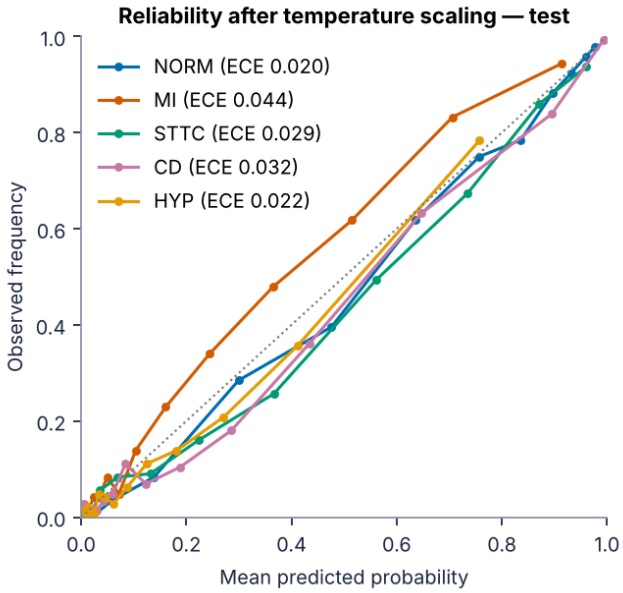
All modelling choices were made on val_select before the freeze. Points are macro-AUROC; bars are patient-bootstrap 95% CIs.

6.2 Internal test (PTB-XL fold 10, single locked run)

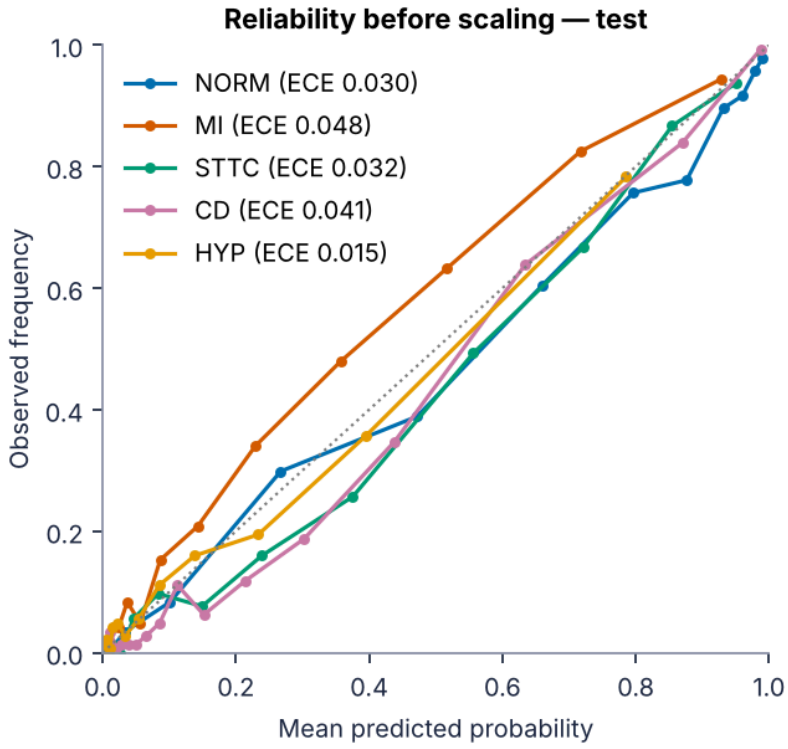
Class	Positives	AUROC [95% CI]	AUPRC	ECE before → after
NORM	963	0.945 [0.935, 0.953]	0.923	0.030 → 0.020
MI	550	0.917 [0.903, 0.930]	0.819	0.048 → 0.044
STTC	521	0.941 [0.929, 0.950]	0.843	0.032 → 0.029
CD	496	0.925 [0.908, 0.939]	0.854	0.041 → 0.032
HYP	262	0.897 [0.875, 0.917]	0.668	0.015 → 0.022



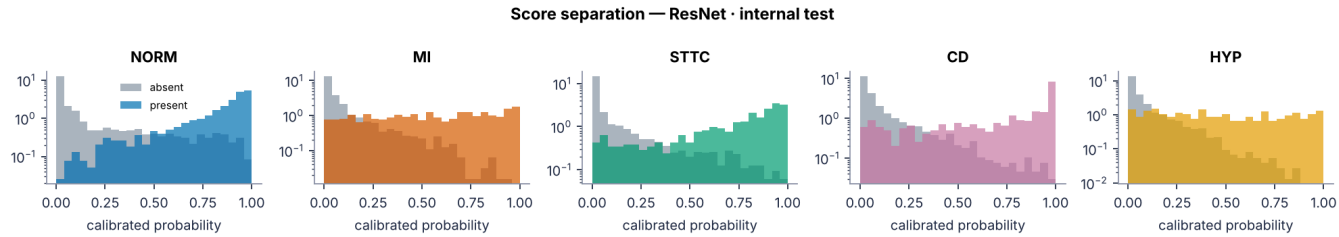
ROC curves on the internal test fold.



Reliability after temperature scaling.



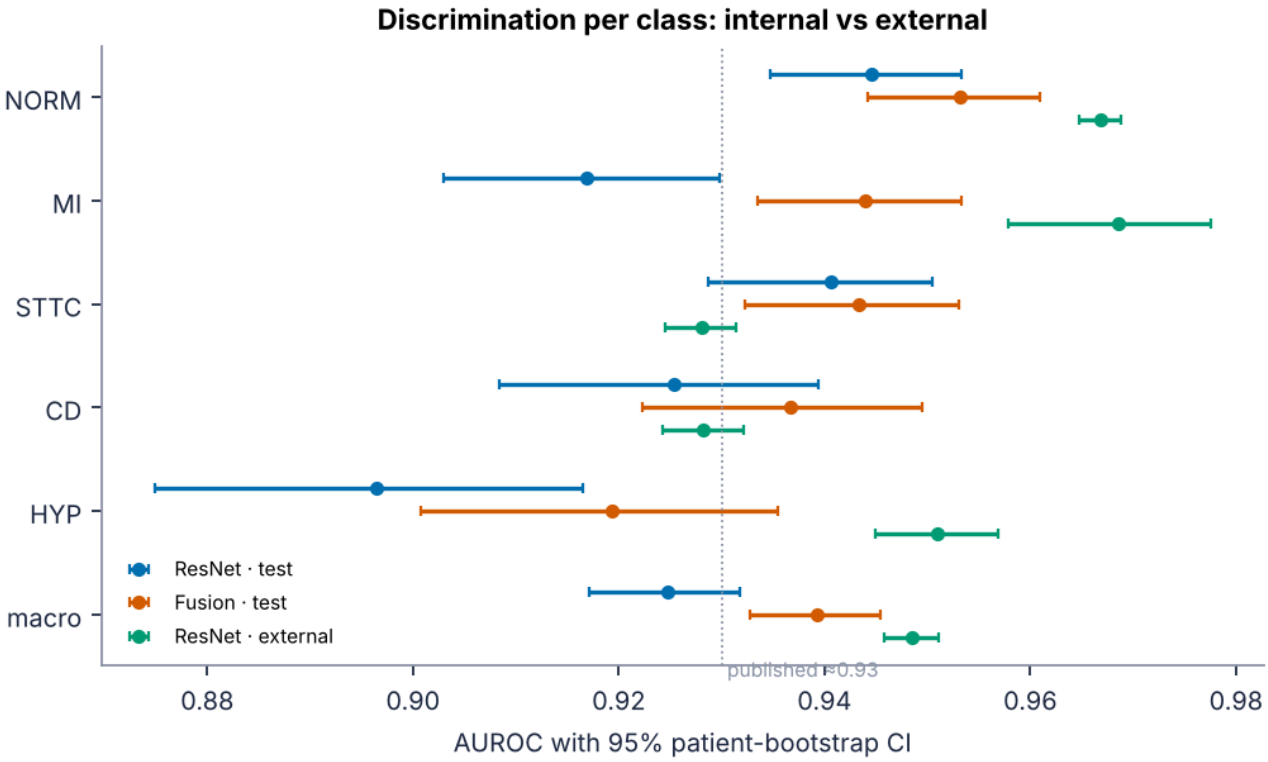
Reliability before scaling, for comparison.



Calibrated probability for ECGs with (colour) and without (grey) each label, internal test. Overlap is where errors and referrals happen.

6.2b ResNet + 12SL fusion (single locked run)

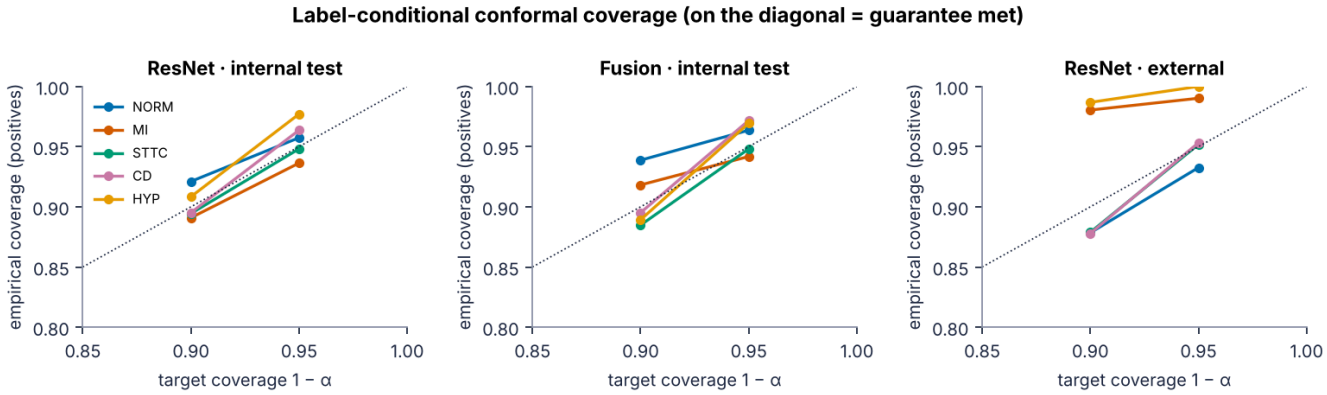
The fused model reaches macro-AUROC 0.939 (95% CI 0.933–0.945) on the same test ECGs, against 0.925 for the ResNet alone, and is at least as good in every class.



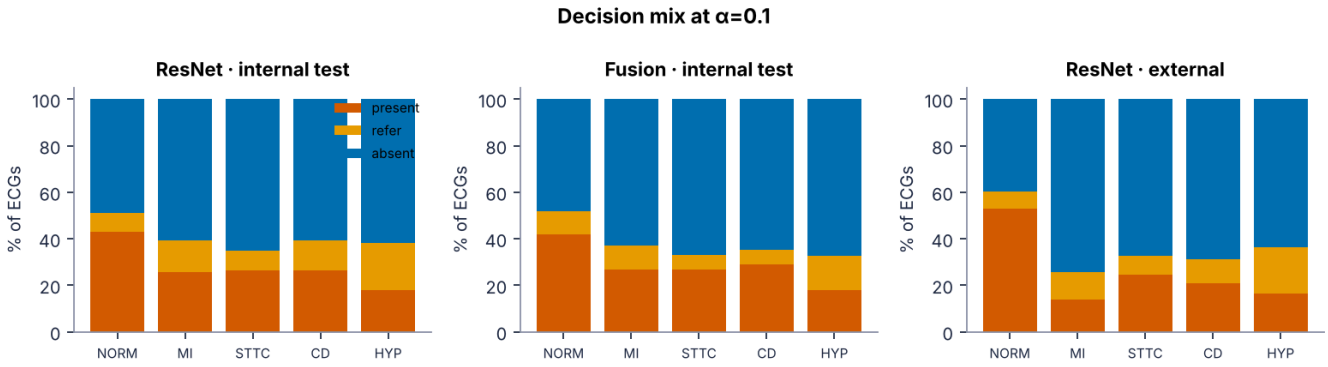
Per-class AUROC with 95% CIs for the three locked evaluations; the dotted line is the published ≈0.93 benchmark.

6.3 Calibrated abstention

α	Mean coverage (positives / negatives)	Mean refer rate	Accuracy on non-referred
0.05	0.957 / 0.948	32.9%	0.923
0.1	0.902 / 0.901	12.7%	0.887

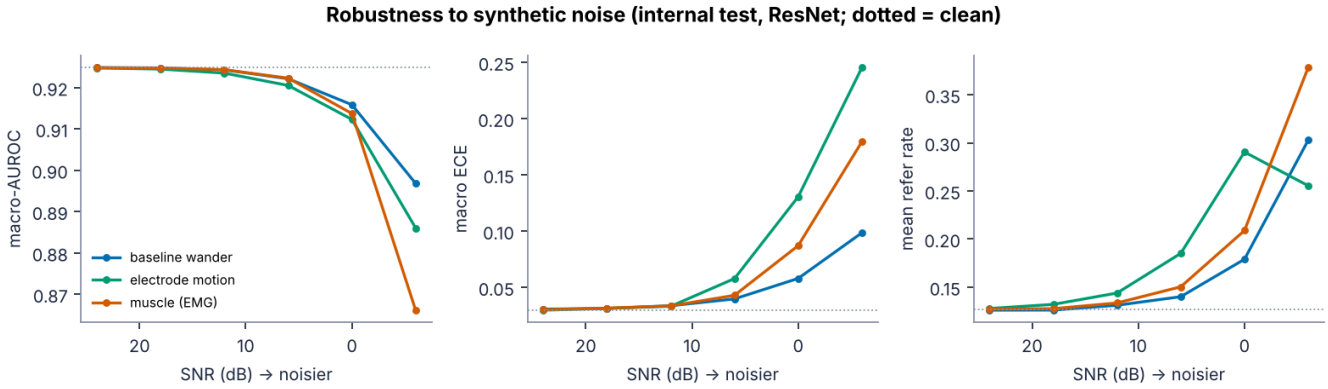


Empirical coverage of true positives vs the 1 − α target, per class. Points on or above the diagonal meet the guarantee; the external panel falls below it for NORM, STTC and CD.

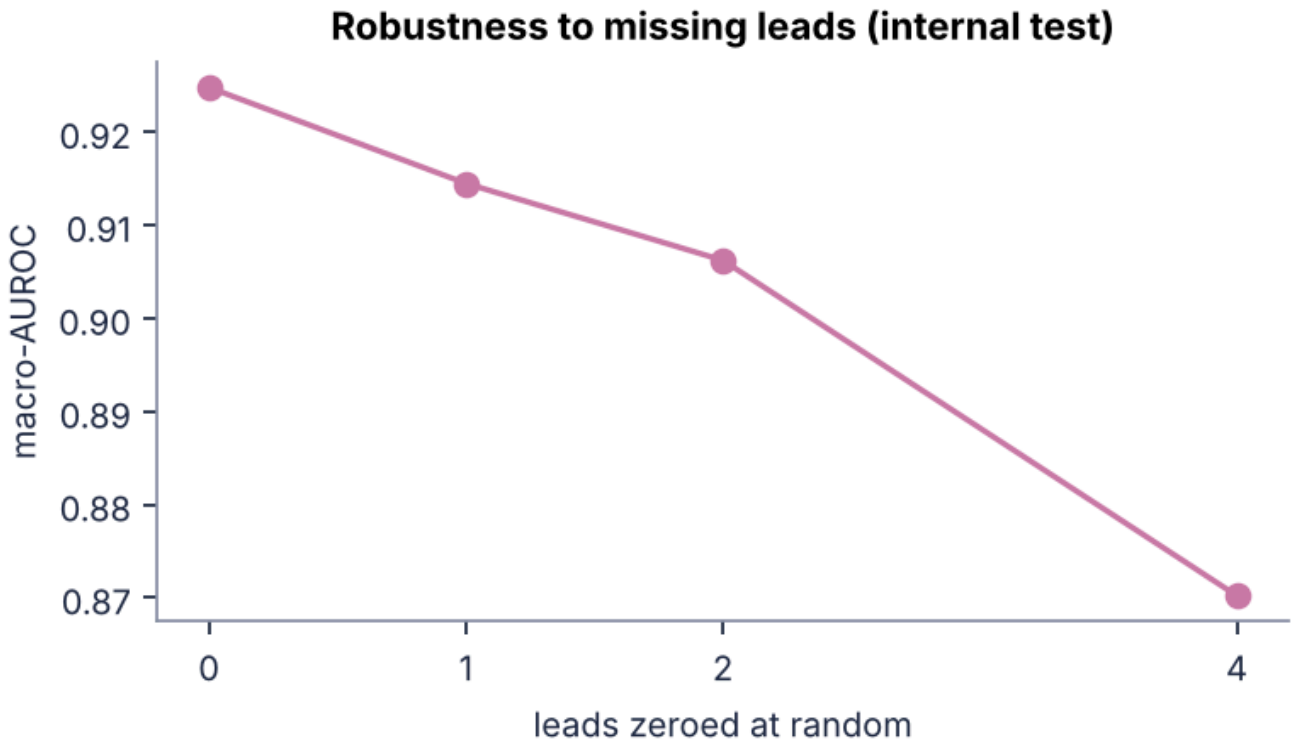


Share of present / refer / absent decisions per class at α = 0.10.

6.4 Robustness to noise

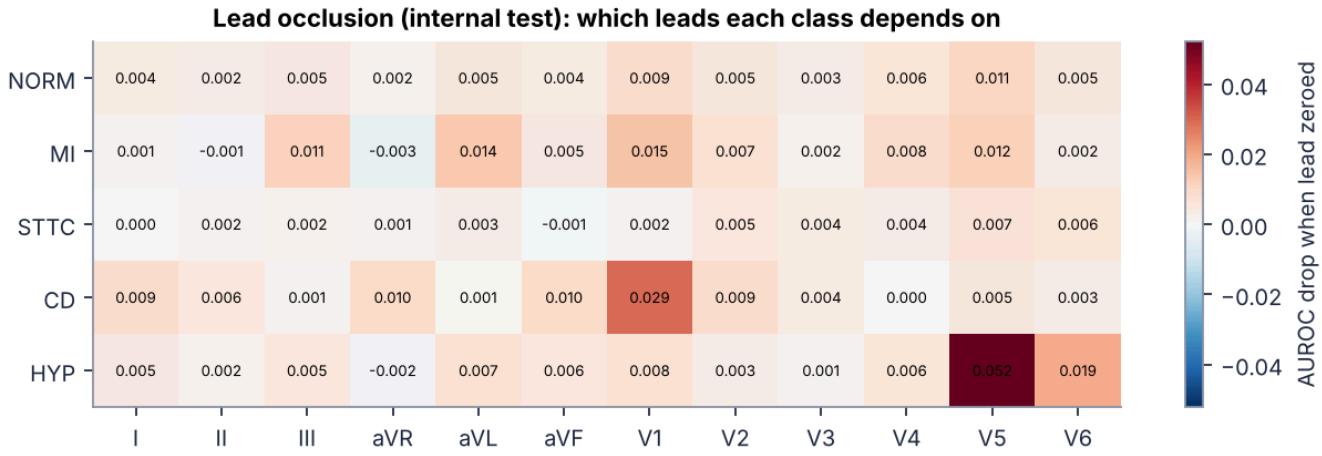


Macro-AUROC, calibration error and refer rate as synthetic noise increases (internal test). At −6 dB electrode motion ECE rises sharply while the refer rate falls: the model becomes confidently wrong rather than abstaining.

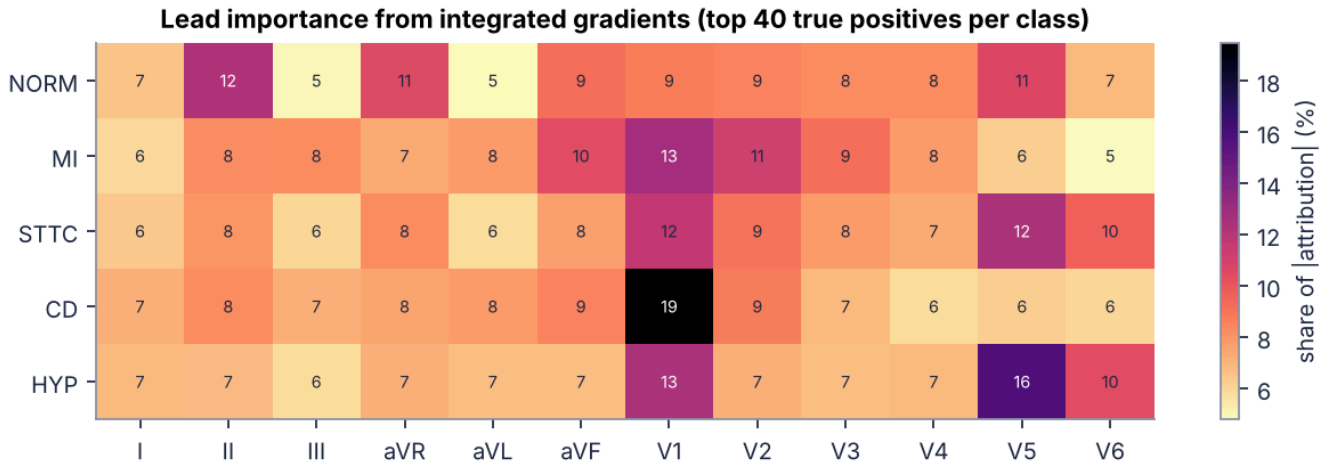


Macro-AUROC when 1, 2 or 4 leads are zeroed at random.

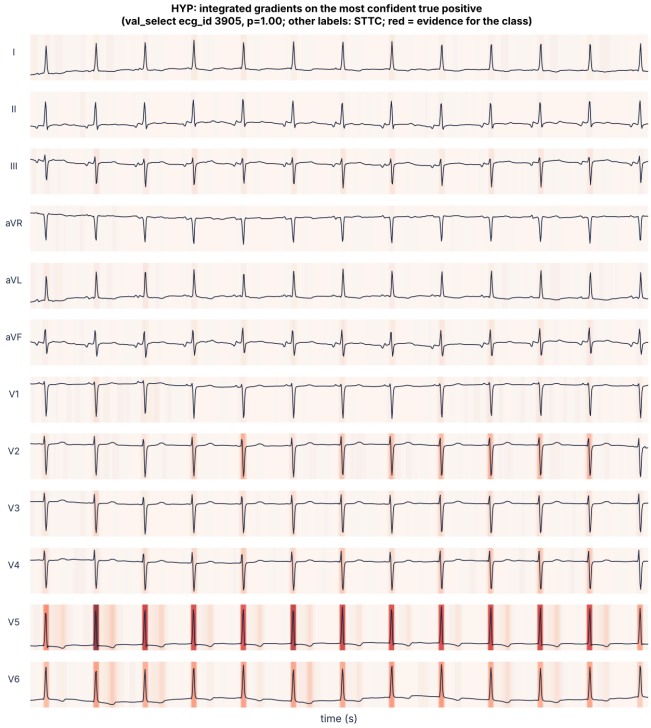
6.5 Explainability: what the model looks at



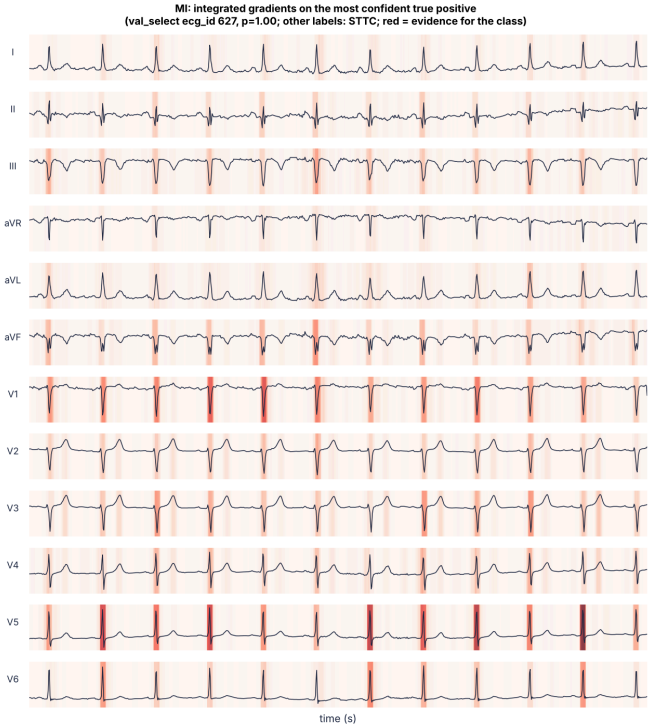
AUROC drop when a single lead is zeroed (internal test). Descriptive only — not a mechanism.



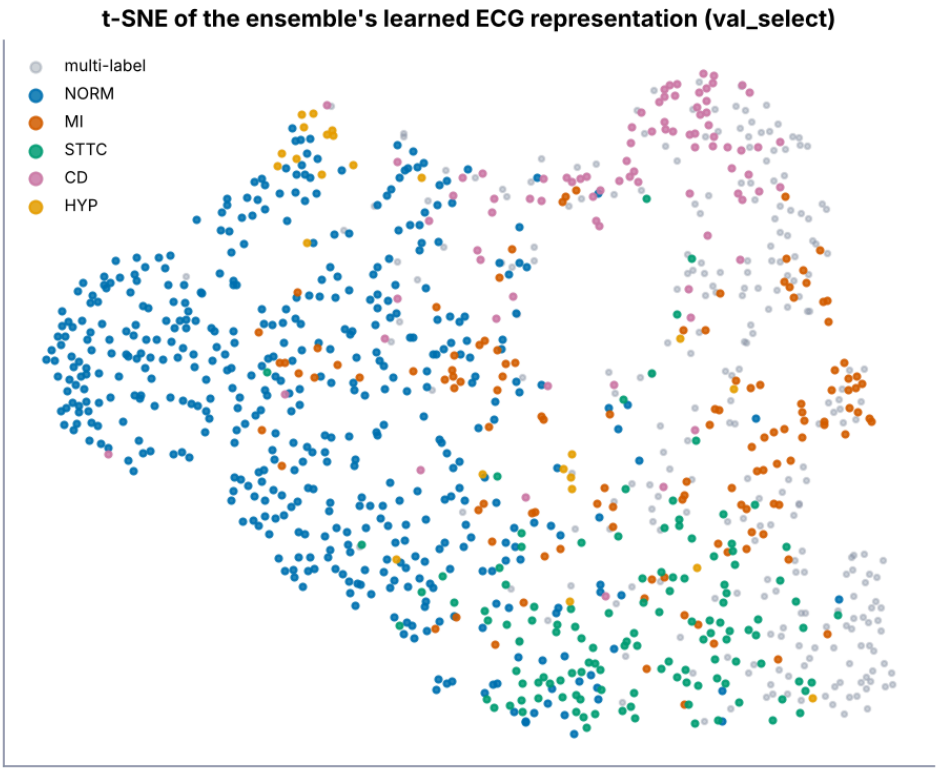
Share of integrated-gradients attribution per lead, averaged over the 40 most confident true positives per class (validation). HYP concentrates on V5, where R-wave voltage criteria are read; CD on V1, where bundle-branch morphology shows.



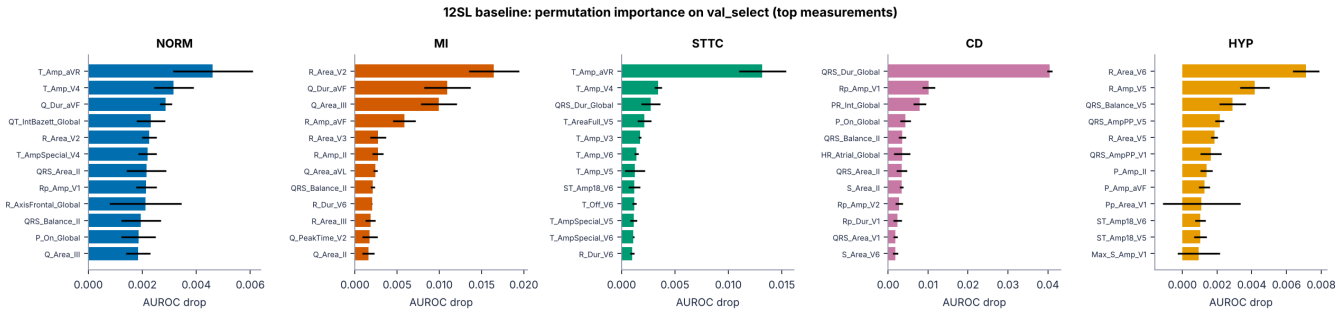
Integrated gradients for HYP on one validation ECG: QRS complexes in V5–V6 carry the evidence.



Integrated gradients for MI on one validation ECG: evidence concentrates on QRS complexes in V1 and V5.

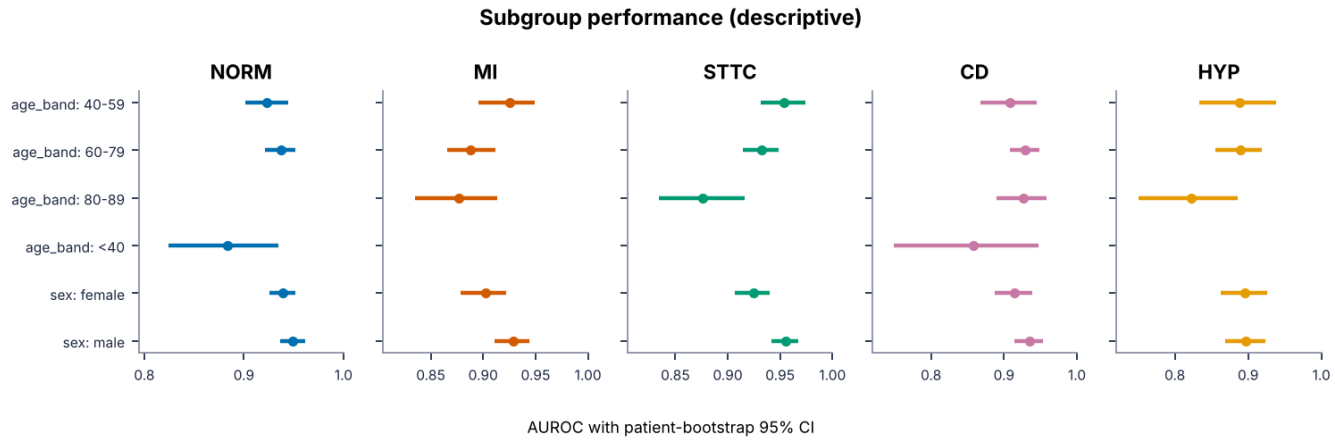


t-SNE of the ensemble's learned representation (validation). Normal ECGs form a separate region; abnormal classes overlap where patients carry several labels.



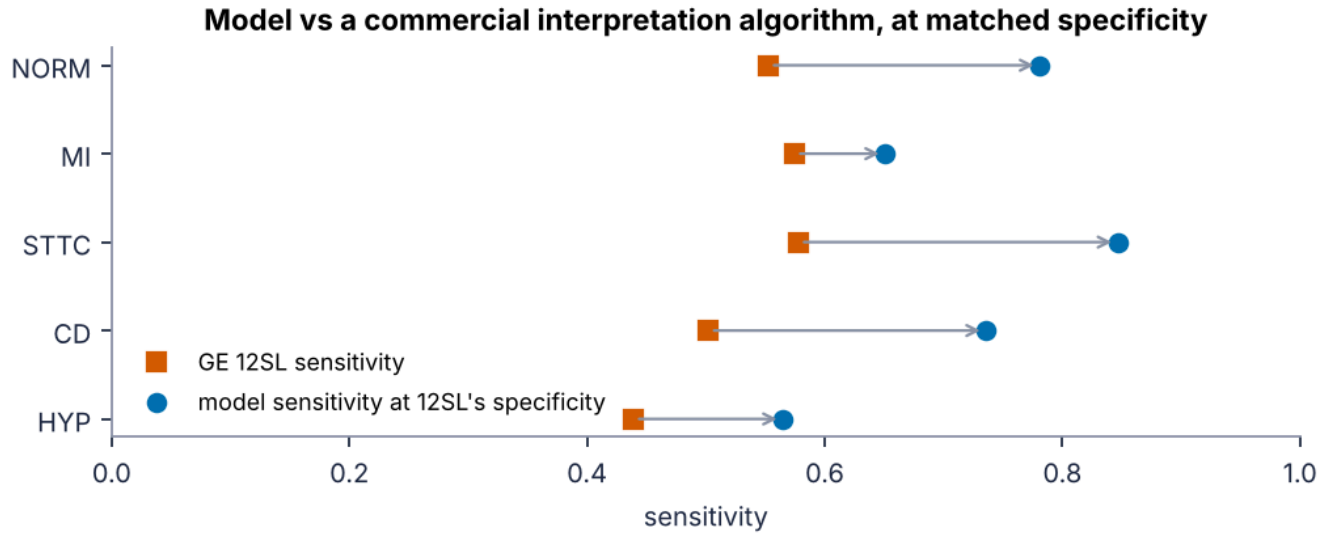
What the 12SL feature model relies on: permutation importance on validation (AUROC drop), top measurements per class.

6.6 Subgroups

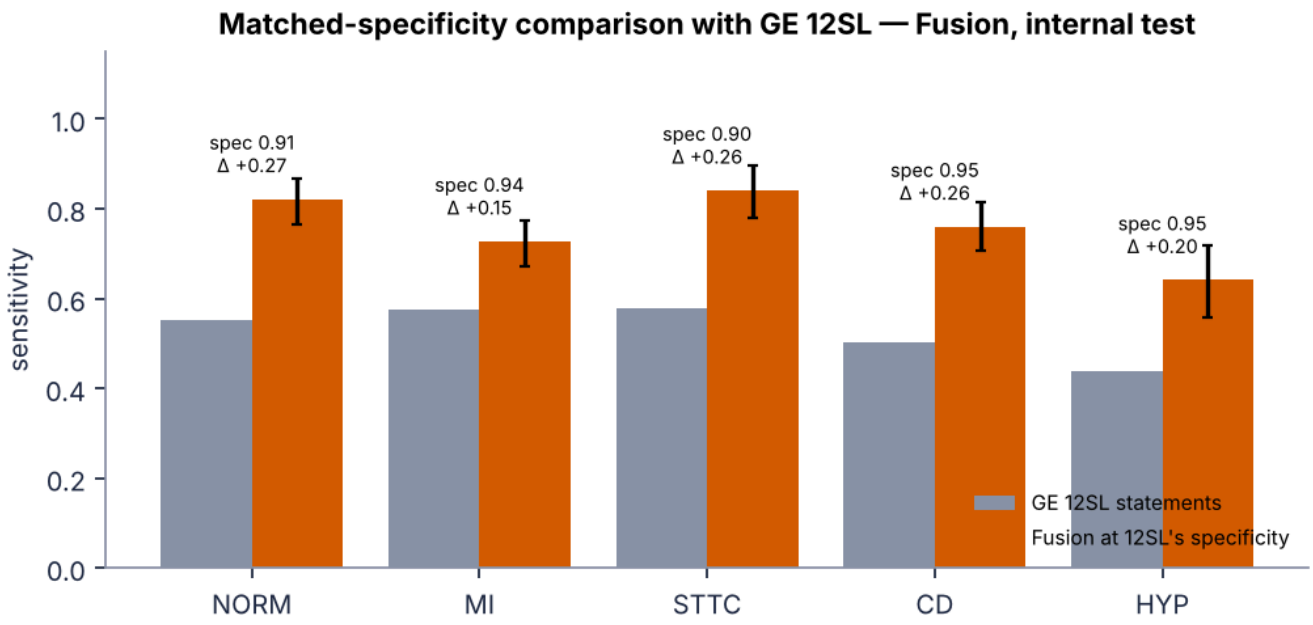


Descriptive subgroup AUROCs with patient-bootstrap 95% CIs (internal test). Differences are only noted when intervals do not overlap.

6.7 Comparison with GE 12SL



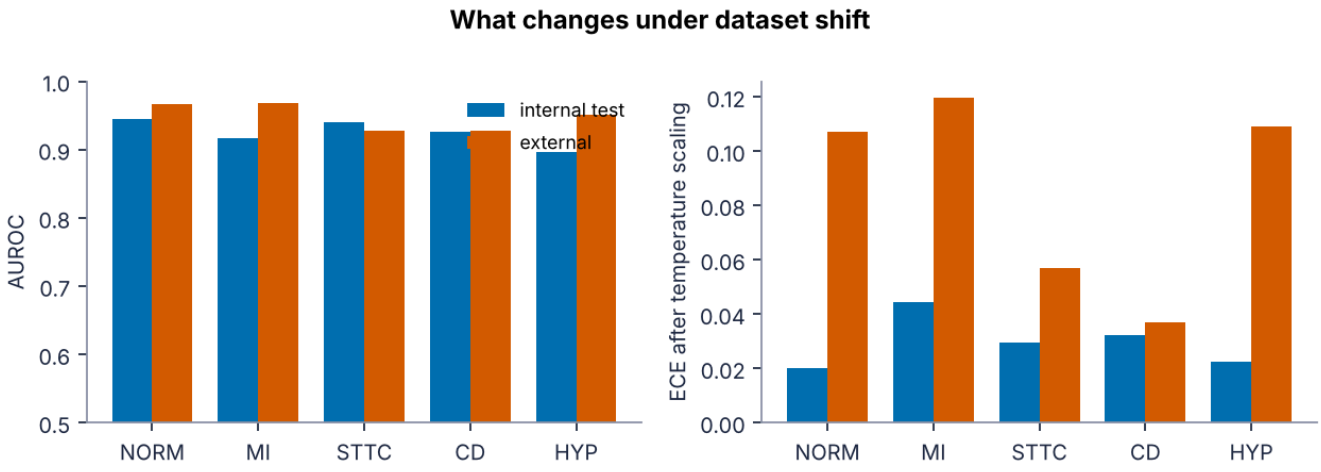
Model sensitivity at 12SL's own specificity on the same test ECGs. Indicative: depends on the published statement map.



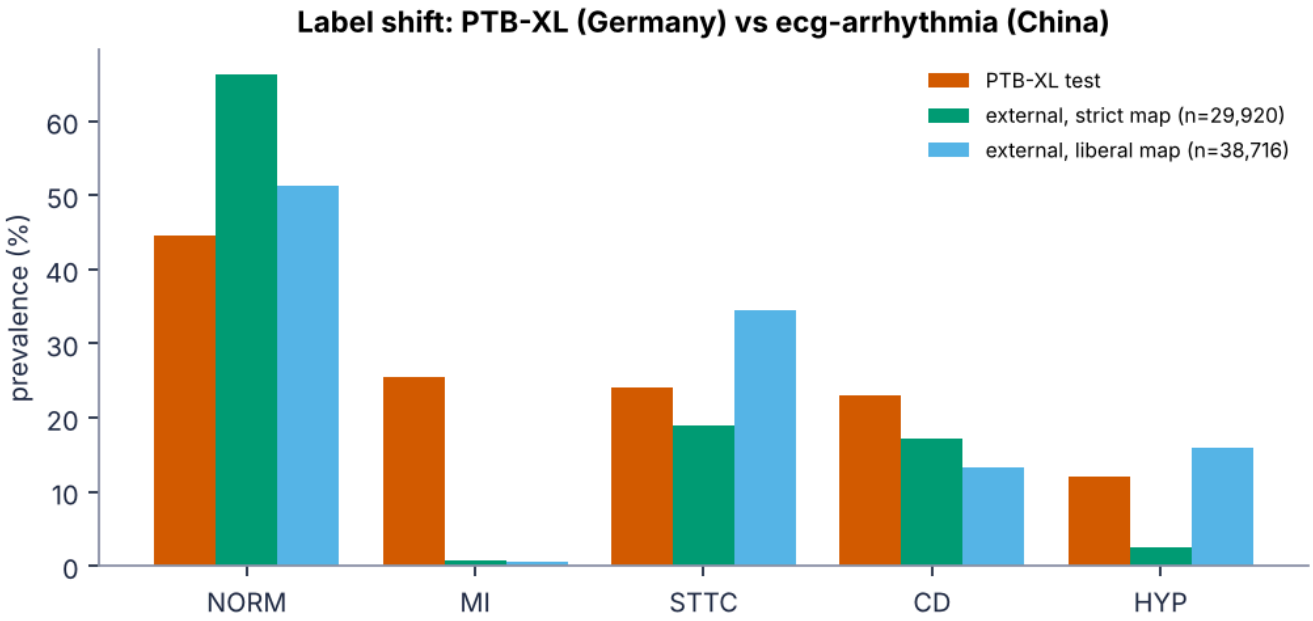
The same comparison for the fused model: every class now gains sensitivity with a CI excluding zero.

6.8 External validation

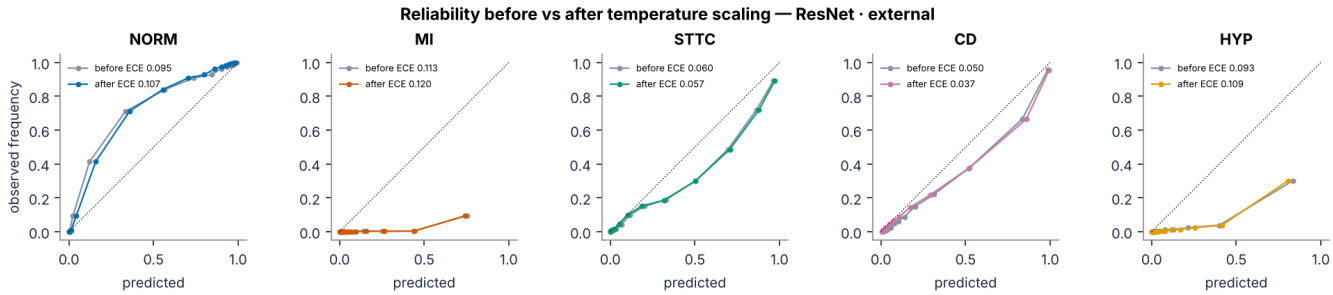
On 28,646 external ECGs the macro-AUROC is 0.949 (95% CI 0.946–0.951). Per-class AUROC changes by -0.054 to +0.013 (internal minus external), and calibration error rises, as expected under label and population shift.



Per-class AUROC and calibration error, internal test vs external validation.



Label shift between PTB-XL and the external cohort: under the strict map MI is rare (0.7%) and NORM dominates, which makes the external task easier and its AUROC not comparable to the internal one.



Reliability on the external cohort: temperatures fitted on PTB-XL do not transfer, so probabilities are miscalibrated under shift.

7 · Limitations and non-claims

- **Not a medical device.** The model is research software, not clinically validated, and must not be used for diagnosis.
- **Noisy reference standard.** PTB-XL labels come from reports; about a third of training labels were not validated by a cardiologist.
- **Marginal guarantees.** Conformal coverage holds on average under exchangeability, not for every patient, and not under dataset shift.
- **Descriptive subgroups.** Many subgroups are compared; small groups are flagged and differences are not causal.
- **Label harmonisation.** External labels are mapped, and the NORM proxy is imperfect; the sensitivity map shows how much this matters.
- **Fusion is PTB-XL only.** It needs GE 12SL measurements; the served model and the external evaluation are ResNet-only.
- **Shift breaks the guarantees.** On the external cohort calibration error roughly triples and positive coverage at $\alpha = 0.10$ falls to about 0.88 for three classes; local re-calibration is required.
- **Severe noise.** At −6 dB electrode motion the model is over-confident instead of referring; a signal-quality gate is needed upstream.
- **Synthetic noise.** The real NSTDB noise records were not reachable from the compute environment; synthetic noise is parameterised to mimic them.
- **Comparator.** The 12SL comparison depends on a statement map and on 12SL's binary output; it is indicative.

8 · Reproducibility

```
pip install torch --index-url https://download.pytorch.org/whl/cpu
pip install -e ".[serve,dev]"
ecgtrust download --dataset ptbxl && ecgtrust download --dataset ptbxl_plus && ecgtrust download
--dataset external
ecgtrust prepare                       # splits + 100 Hz caches
ecgtrust audit
ecgtrust train --model baseline --features 12sl
ecgtrust train --model resnet --seeds 0 1 2
ecgtrust train --model fusion
ecgtrust evaluate --split val_select
ecgtrust freeze                        # hashes checkpoints, config and label maps
ecgtrust evaluate --split test         # single use, logged
ecgtrust evaluate --split test --model fusion
ecgtrust evaluate --split external     # single use, logged
ecgtrust results && ecgtrust figures && ecgtrust release
ecgtrust guide                         # rebuilds this PDF from reports/
```

Freeze manifest: `aa872cf43f42`. Seeds 0, 1, 2. Python 3.13.18, PyTorch 2.15.0. Every unit test runs in CI on synthetic fixtures without network access.

9 · References

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